

Object Recognition on CIFAR-100: A Comparative Study of Generative, Hand-Crafted, and Deep Learning Approaches

1. Introduction

Object recognition remains a central challenge in computer vision, particularly when models must distinguish between visually similar object categories. This challenge is amplified in fine-grained settings, where intra-class variation and inter-class similarity can overwhelm low-level visual cues.

This work investigates object classification on the CIFAR-100 dataset (Krizhevsky, 2009), which contains 100 fine-grained categories grouped into 20 coarse super-classes, with balanced class distributions across both levels. This work compares three increasingly expressive approaches:

- PCA-enhanced Gaussian Mixture Model (GMM): an unsupervised baseline that highlights the limitations of clustering raw pixel statistics.
- Histogram of Oriented Gradients (HOG) + Support Vector Machine (SVM): a classical pipeline using hand-crafted features, providing a reference for the improvements gained by deep learning.
- Convolutional Neural Network (CNN): an end-to-end hierarchical feature learning model capable of capturing both low-level patterns (edges, textures) and high-level semantic representations, making it well-suited for hierarchical datasets like CIFAR-100.

These methods are evaluated using overall accuracy (general performance), per-class accuracy (weaknesses on specific categories), confusion matrices (patterns of errors), and sub-region analysis, focusing on the “large carnivores” super-class to study fine-grained discrimination within a semantically coherent subset. The CNN achieves over 41% fine-label accuracy and 54.8% coarse-label accuracy, substantially outperforming classical baselines and exceeding the provided benchmark.

2. Methods

2.1 Dataset and Exploratory Analysis

The CIFAR-100 dataset (Krizhevsky, 2009) has 50,000 training images and 10,000 test images, each 32×32 pixels with 3 colour channels. It contains 100 fine-grained categories grouped into 20 coarse super-classes, with exactly 500 images per fine class and 2,500 images per super-class. This perfect balance ensures that differences in performance reflect model capability rather than data imbalance. All images are converted to floating-point numbers and normalized before training or feature extraction.



Figure 1 raw data from the CIFAR-100 dataset with their relevant fine classification.

This work focuses in on the “large carnivores” super-class, which includes visually similar animals like bears, leopards, lions, tigers, and wolves. This subset is particularly challenging because of high inter-class similarity and low image resolution (Figure 1). Confusion matrices for this sub-region are analysed for all methods to understand specific error patterns. Models are trained on the fixed training split and evaluated once on the test set using overall accuracy, per-class accuracy, and confusion matrices. No cross-validation is performed, but the dataset size and balance make this evaluation meaningful.

2.2 Gaussian Mixture Model (GMM) Baseline with PCA

For a low-level baseline, each image is flattened into a 3,072-dimensional vector. Principal Component Analysis (PCA) reduces this to 100 dimensions, chosen to retain approximately 95% of the variance while keeping the model

computationally feasible. A Gaussian Mixture Model with 100 components is fitted to the reduced features, and each component is assigned to a fine class by majority voting over the training labels.

This unsupervised approach tests how well global pixel statistics can separate classes without supervision and allows us to establish a baseline for fine-grained classification performance. The limitations of this approach, particularly on low-resolution and visually similar categories like large carnivores, are evident in the confusion matrix (Figure 4).

2.3 Histogram of Oriented Gradients (HOG) Features with SVM

Next, HOG features (Dalal and Triggs, 2005) are computed with 4x4 pixels per cell and 2x2 cells per block, producing a 1,764-dimensional feature vector per image. This configuration balances the small image size against the need to capture local edge and texture information. A linear Support Vector Machine (SVM) (Cortes and Vapnik, 1995) is trained on these features, with the regularization parameter C chosen via grid search to optimise classification performance while avoiding overfitting.

Visualization of HOG features (Figure 2) demonstrates how local gradients encode shape and texture information. By comparing HOG+SVM to GMM, the improvement obtained from structured, hand-crafted features can be isolated.

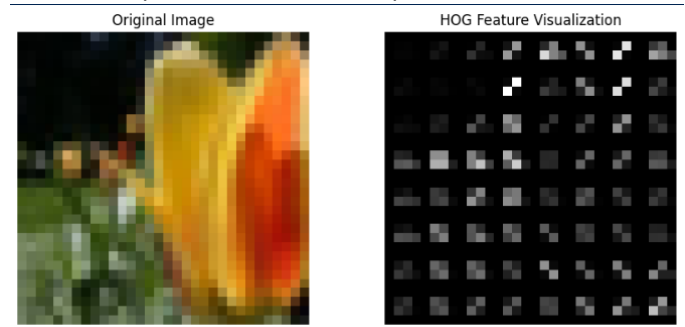


Figure 2 visualisation of HOG feature extraction

2.4 Convolutional Neural Network (CNN)

The CNN is trained directly on raw pixels, learning hierarchical features end-to-end, following the principles of gradient-based learning for visual recognition (LeCun et al., 1998). The network architecture varies in depth (2-4 convolutional layers), filter size (3x3 to 5x5), and number of filters per layer (32-128). A small grid search over 12 configurations identifies the best-performing architecture. Shallower networks train faster but may underfit, while deeper networks capture more semantic patterns at the risk of overfitting.

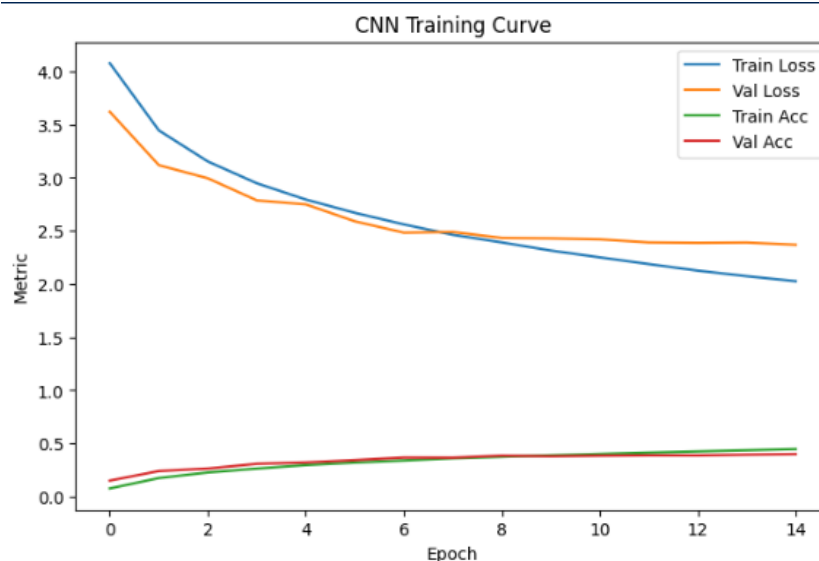


Figure 3 showing the training curve of the convolutional neural network trained on CIFAR-100 data

Training uses the Adam optimizer (Kingma and Ba, 2015) with categorical cross-entropy loss for 15 epochs. Training and validation curves (Figure 3) show smooth convergence with no overfitting. A second CNN trained on coarse labels (20 classes) enables direct comparison between fine- and coarse-level recognition performance. Confusion matrices for the large carnivores (Figure 4) reveal strong diagonal structure, particularly for wolves, while remaining errors are largely confined to visually similar species such as lions and tigers.

2.5 Comparative Methodology and Reasoning

These three methods (GMM, HOG+SVM, and CNN) were chosen to show increasing model expressivity. The GMM tests unsupervised global pixel clustering, HOG+SVM tests supervised hand-crafted features, and CNNs learn hierarchical representations directly from raw pixels. This progression allows us to isolate how each level of model complexity impacts performance, particularly for fine-grained challenges like distinguishing large carnivores.

3. Results

3.1 Overall Quantitative Performance

Method	Fine Accuracy	Sub-region Accuracy	Mean Coarse Accuracy
GMM	0.0894	0.058	0.0917
HOG + SVM	0.1935	0.094	0.1969
CNN	0.4107	0.304	0.4162

Table 1 summarises the performance of all methods across fine-label accuracy, mean coarse accuracy, and sub-region accuracy for the “large carnivores” super-class.

The GMM baseline achieves only 8.9% fine-label accuracy and fails on nearly half of the classes. HOG+SVM more than doubles performance (19.3%) but still struggles with subtle distinctions, as seen in Figure 4. The CNN significantly outperforms classical approaches with 41% fine-label accuracy and 54.8% coarse-label accuracy, exceeding the provided benchmark. Training the CNN on coarse labels demonstrates the strong effect of reducing class ambiguity, improving mean coarse accuracy.

3.2 Per-Class and Coarse-Class Analysis

Detailed per-class results (Table 1) reveal that the GMM baseline has zero accuracy on 48 classes, while HOG+SVM achieves non-zero accuracy for all classes but struggles with animals such as beavers, seals, and aquatic mammals. The CNN improves worst-case performance, although small mammals and aquatic species remain challenging. Figure 4 illustrates per-class confusion patterns, highlighting the diagonal dominance of the CNN relative to GMM and HOG+SVM.

Mean accuracy per coarse class consistently increases with model complexity (Table 2), showing that deep representations enable strong performance on mammals, non-insect invertebrates, and large natural scenes.

3.3 Confusion Matrix and Sub-region Analysis

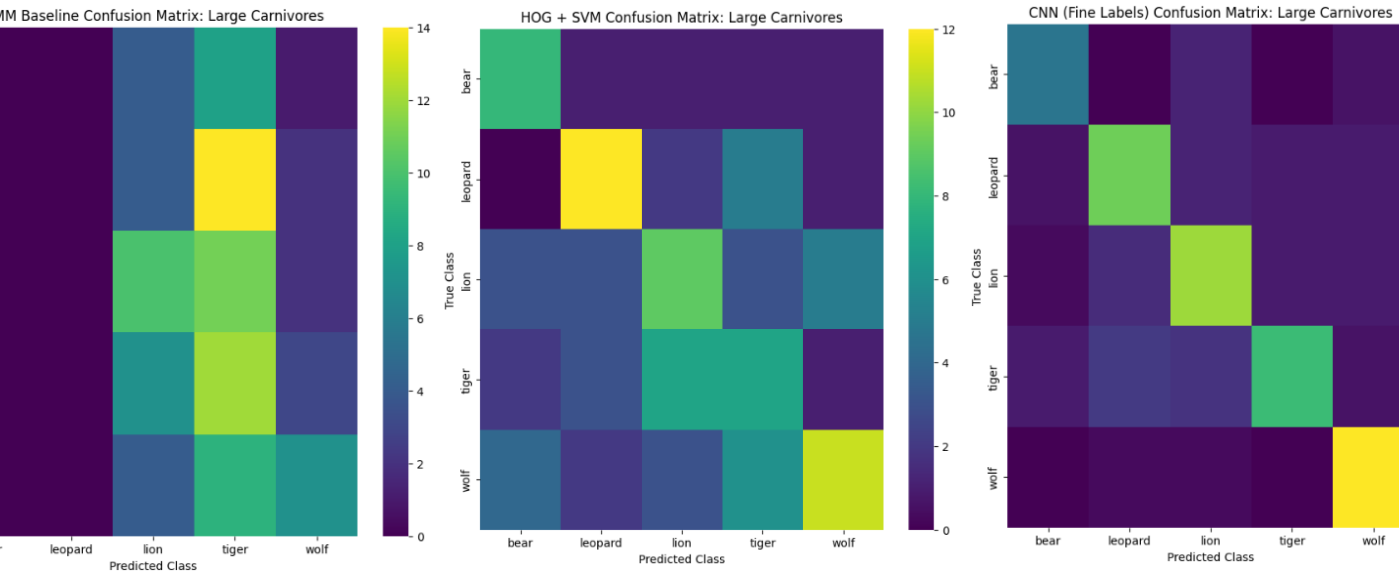


Figure 4 all three methods analysed on a single superclass “Large Carnivores”

Figure 4 clearly demonstrates the differences in error structure across models. GMM shows almost no diagonal structure, with widespread misclassification across large carnivore classes. HOG+SVM exhibits modest improvement, with partial diagonal formation and better recognition of structurally distinct species such as bears and leopards. The CNN confusion matrix shows strong diagonal dominance, particularly for wolves, while remaining errors occur among visually similar

Class Name	GMM	SVM	CNN	Mean Accuracy
seal	0.00	0.01	0.07	0.026667
otter	0.00	0.04	0.05	0.030000
squirrel	0.00	0.07	0.08	0.050000
beaver	0.00	0.01	0.15	0.053333
snail	0.00	0.05	0.14	0.063333
possum	0.07	0.02	0.15	0.080000
bear	0.00	0.08	0.16	0.080000
mouse	0.00	0.08	0.17	0.083333
raccoon	0.00	0.06	0.20	0.086667
lobster	0.00	0.09	0.18	0.090000

Table 2 per fine class breakdown of accuracy for the three models

species like lions and tigers. Sub-region analysis (large carnivores) highlights limitations of fine-grained recognition: GMM achieves only 5.8%, HOG+SVM 9.4%, and CNN 30.4%, despite correctly classifying wolves with >95% accuracy. These figures (Figures 1, 2, 4) illustrate that even deep models are constrained by low resolution and semantic similarity.

4. Conclusion

This study compared unsupervised, classical, and deep learning approaches for object recognition on CIFAR-100. Results show that increasing representational capacity not only improves accuracy but fundamentally changes the structure of classification errors, shifting confusion from cross-category failures to within-class semantic ambiguity.

However, detailed per-class and sub-region analysis (Figure 4) reveals persistent failure modes, particularly for visually similar animal categories, indicating that limited image resolution and semantic overlap fundamentally constrain performance.

While the results demonstrate a clear progression in performance from unsupervised to deep learning methods, several limitations should be acknowledged. All models are evaluated using a single fixed train-test split, without repeated runs or cross-validation, meaning variability due to random initialisation (particularly for CNN training) is not explicitly measured. Additionally, the low spatial resolution of CIFAR-100 images (32×32 pixels) and the presence of visually ambiguous fine-grained classes, such as those within the large carnivores super-class, inherently limit the achievable accuracy, even for deep models.

Future work could address these limitations in several ways. Repeated training runs or cross-validation would provide more robust estimates of performance and variance. As shown in Figure 3, the CNN converges stably, suggesting that further gains are more likely to come from richer input representations rather than extended training. Hierarchical learning strategies, such as incorporating coarse labels into the fine-label loss function or adopting multi-task learning, could better exploit the structured label space of CIFAR-100. Additionally, transfer learning from higher-resolution datasets and stronger data augmentation strategies may help mitigate resolution constraints and improve fine-grained discrimination for visually similar categories.

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